Task 11: Implement Traffic Management Using Azure Traffic Manager

Objective

Implement Azure Traffic Manager to manage incoming traffic between two virtual machines and provide automatic failover using Priority Routing.

Step 1: Verify Existing Resources

Navigate to:

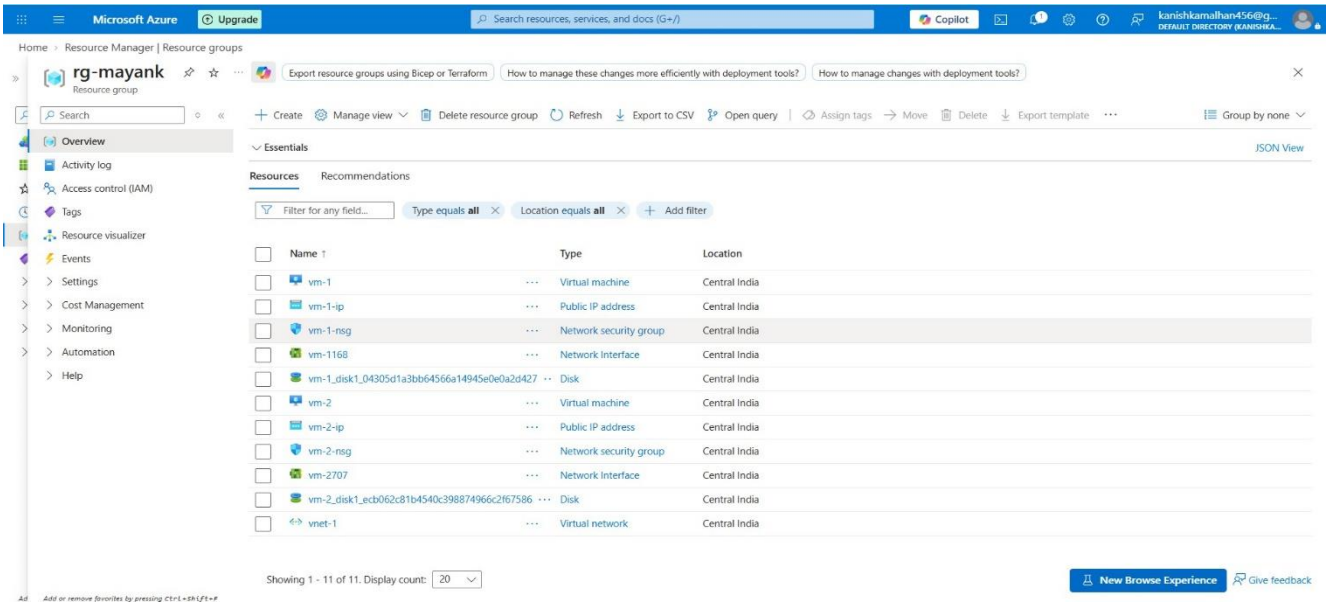
Resource Group → rg-mayank

Verify the following resources are available:

- vnet-1
- vm-1
- vm-2
- vm-1-ip
- vm-2-ip
- Network Security Groups
- Network Interfaces

Observation

All required networking and compute resources were successfully deployed.



Step 2: Deploy StreamFlix Application on VM-1

Connect to VM-1 through SSH.

Clone the application repository:

cd StreamFlix
ls

Observation

The StreamFlix application repository was successfully cloned on VM-1.

```
azureuser1@vm-1: $ git clone https://github.com/devopsinsiders/StreamFlix.git
fatal: could not create work tree dir 'StreamFlix': Permission denied
azureuser1@vm-1: $ sudo git clone https://github.com/devopsinsiders/StreamFlix.git
Cloning into 'StreamFlix'...
remote: Enumerating objects: 143, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 143 (delta 2), reused 0 (delta 0), pack-reused 137 (from 1)
Receiving objects: 100% (143/143), 7.95 MiB | 17.37 MiB/s, done.
Resolving deltas: 100% (8/8), done.
azureuser1@vm-1: $ ls
StreamFlix  bin  bin.usr-is-merged  boot  dev  etc  home  lib  lib.usr-is-merged  lib64  lost+found  media  mnt  opt  proc  root  run  sbin  sbin.usr-is-merged  snap  srv  sys  tmp  usr  var
azureuser1@vm-1: $ cd StreamFlix/
azureuser1@vm-1:/StreamFlix$ ls
README-Linux.md  README.md  assets  avatar.png  favicon.ico  index.html  logo192.png  logo512.png  manifest.json  robots.txt
azureuser1@vm-1:/StreamFlix$
```

Step 3: Configure Web Root on VM-1

Copy application files to the Nginx web root directory:

sudo cp -r * /var/www/html/

Verify the copied files:

cd /var/www/html/
ls

Observation

Application files were successfully copied to the web root directory and made available through Nginx.



Step 4: Deploy Starbucks Clone Application on VM-2

Connect to VM-2 through SSH.

Navigate to the application directory:

cd starbucks-clone

ls

Copy application files to the web root directory:

sudo cp -r * /var/www/html/

Verify deployment:

cd /var/www/html/
ls

Observation

The Starbucks Clone application was successfully deployed on VM-2.



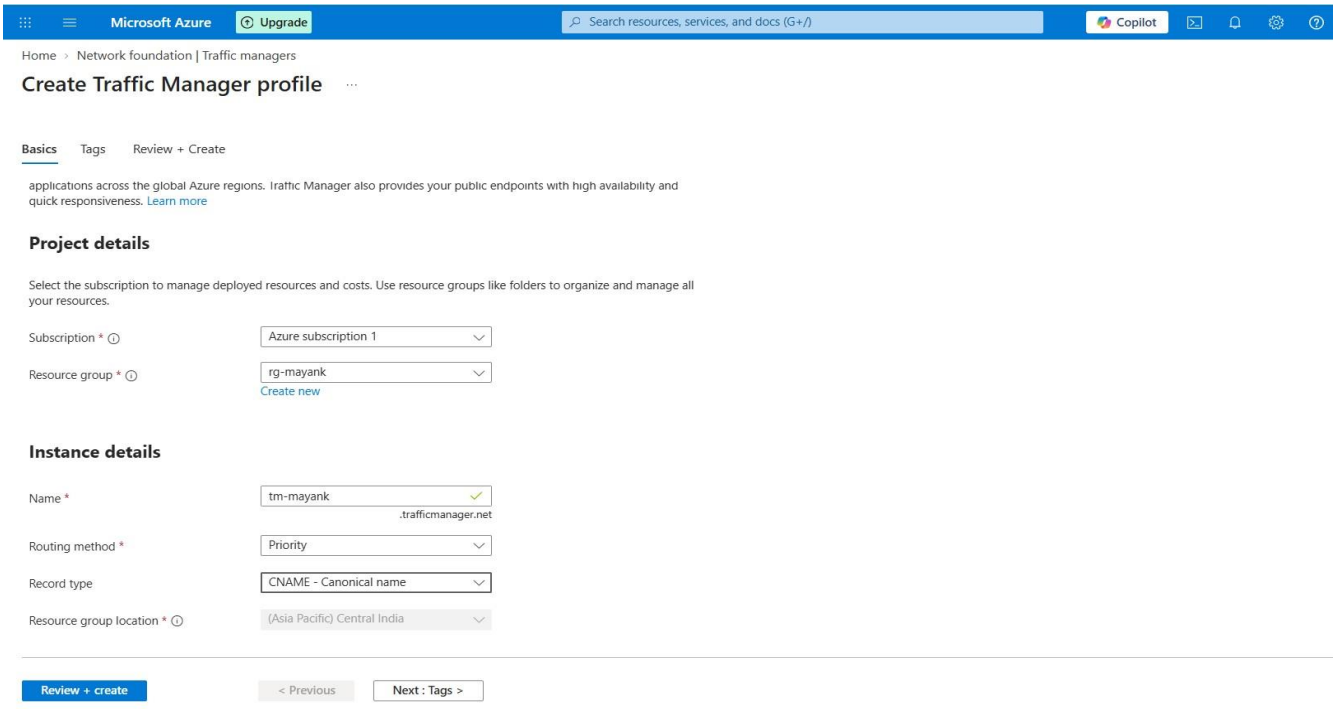
Step 5: Create Traffic Manager Profile

Navigate to:

Azure Portal
→ Traffic Managers
→ Create

Observation

The Traffic Manager profile was successfully created.



Step 6: Configure DNS Labels for Public IP Addresses

Navigate to:

Public IP Addresses

Configure DNS labels:

VM-1

vm1mayank

VM-2

vm2mayank

Save the configuration.

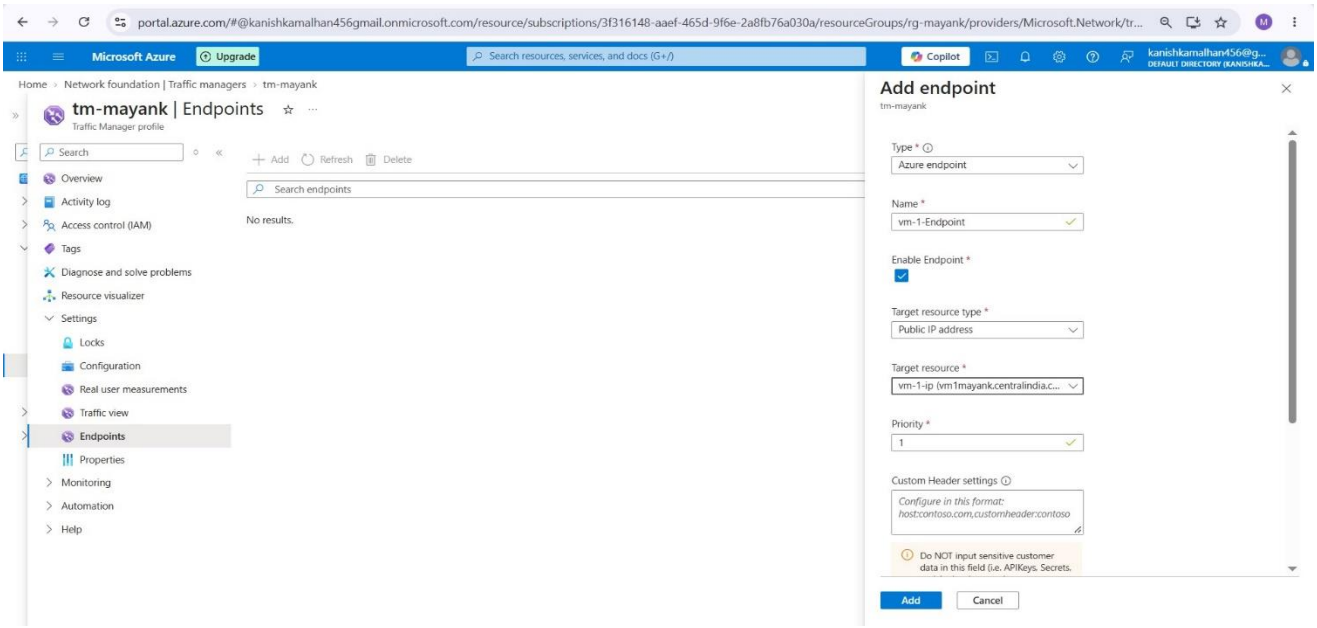
Observation

Public DNS names were generated for both virtual machines.

Step 7: Configure Primary Endpoint

Navigate to:

Traffic Manager
→ Endpoints
→ Add



Step 8: Configure Secondary Endpoint

Navigate to:

Traffic Manager
→ Endpoints
→ Add

Observation

Both endpoints were successfully added to the Traffic Manager profile.

Step 9: Verify Endpoint Health

Navigate to:

Traffic Manager
→ Endpoints

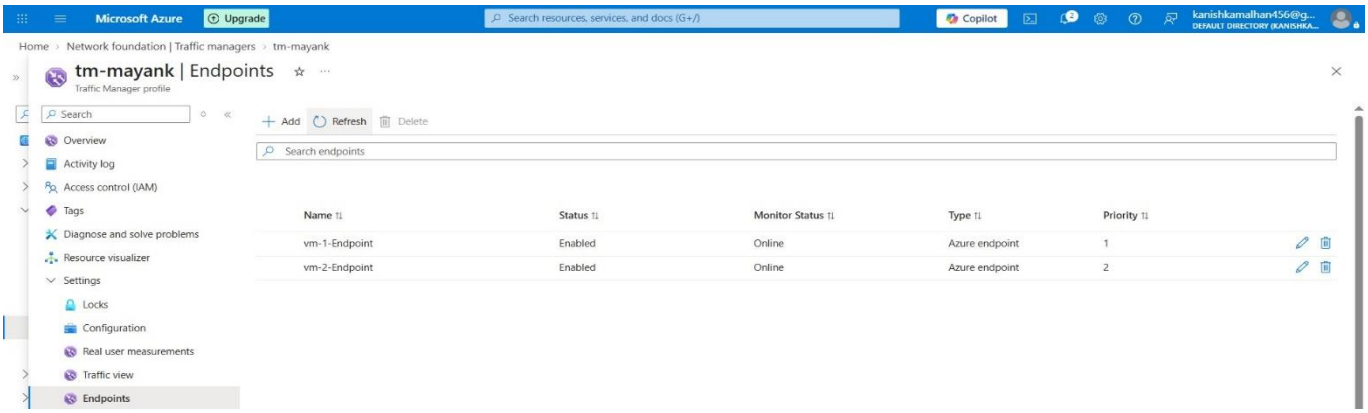
Verify both endpoints display:

Enabled

Online

Observation

Azure Traffic Manager successfully completed health checks and marked both endpoints online.



Step 10: Verify Traffic Routing

Navigate to:

Traffic Manager
→ Overview

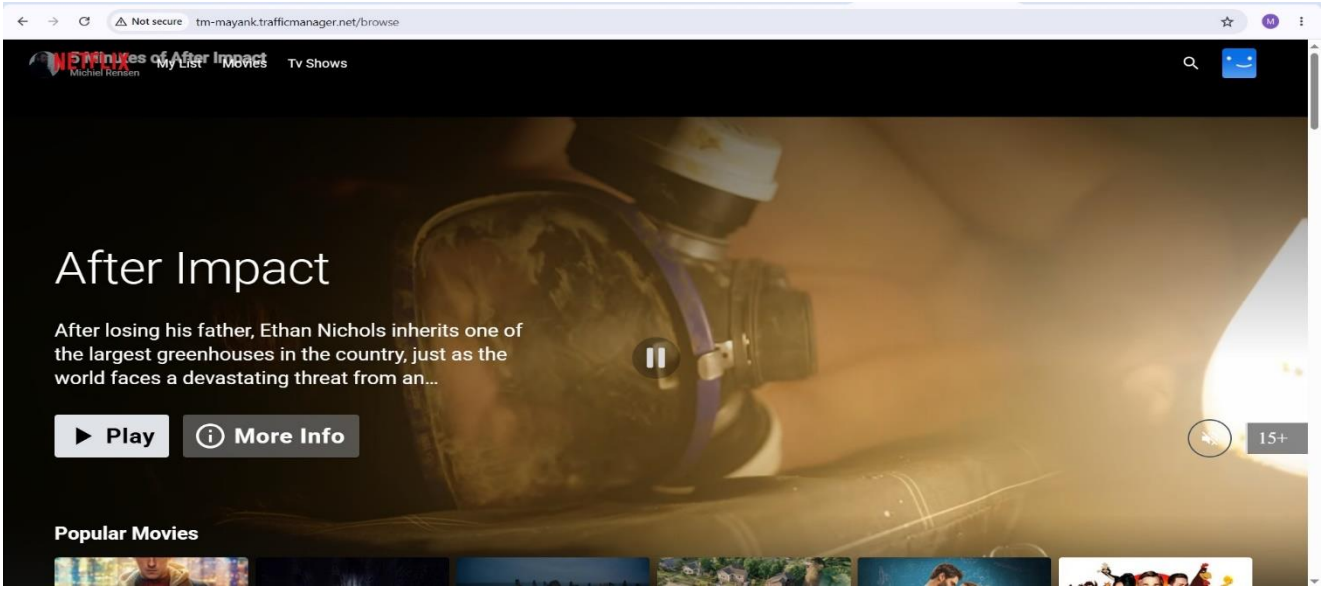
Copy the generated DNS name:

tm-mayank.trafficmanager.net

Open the URL in a web browser.

Observation

Traffic was routed to VM-1 because it was configured as the primary endpoint with Priority 1.



Step 11: Simulate Endpoint Failure

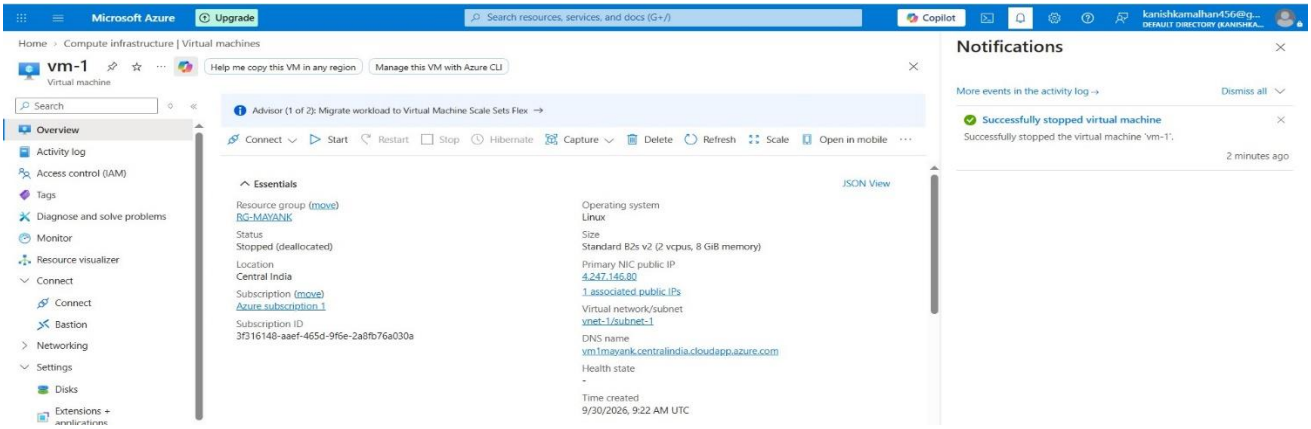
Navigate to:

Virtual Machines
→ vm-1
→ Stop

Wait for Azure Traffic Manager health monitoring to detect the endpoint failure.

Observation

The primary virtual machine was successfully stopped.



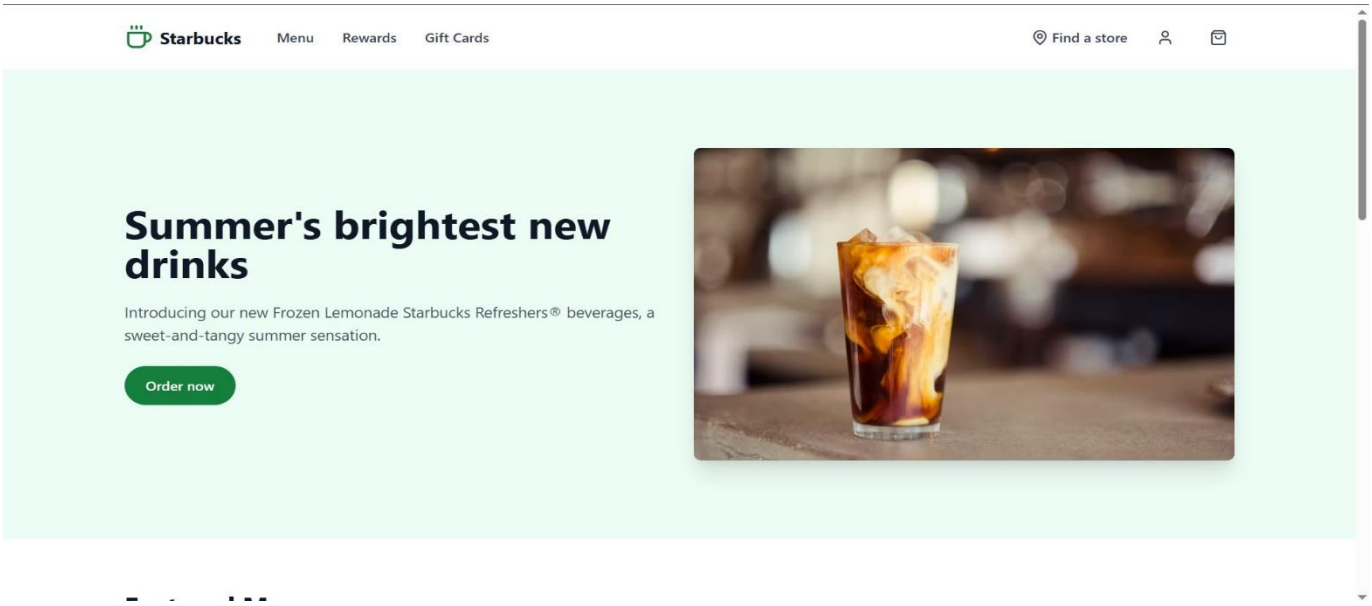
Step 12: Verify Automatic Failover

Access the same Traffic Manager DNS URL again:

http://tm-mayank.trafficmanager.net

Observation

Traffic was automatically redirected to VM-2 and the Starbucks Clone application was displayed, confirming successful failover to the secondary endpoint.



Result

Successfully implemented Azure Traffic Manager using the Priority Routing Method. Traffic was initially routed to the primary endpoint (VM-1). After the primary endpoint became unavailable, Azure Traffic Manager automatically redirected user requests to the secondary endpoint (VM-2), ensuring application availability and fault tolerance.